
Ponderomotive nonlinearity in a spin-chain emulator of plasma waves: a ZZ -coupled model, its nonlinear-Schrödinger limit, and a full-Hilbert-space test

Anonymous

Abstract

Spin chains whose single-excitation spectrum reproduces the electromagnetic dispersion relation of an unmagnetised plasma give near-term quantum hardware a shallow-depth route to full-wave plasma simulation, but only in the linear regime, where the model is a free lattice field theory and is classically simulable in polynomial time. We show that adding a uniform nearest-neighbour $\sigma_i^z \sigma_j^z$ coupling to such a chain generates, exactly and with no further assumptions, the *ponderomotive* nonlinearity of plasma physics: the local plasma frequency acquires a term proportional to the local wave intensity, with a coefficient we compute in closed form. Reducing the model with a spin-coherent (classical-field) variational principle yields a discrete nonlinear Dirac system whose envelope obeys a cubic nonlinear Schrödinger equation with nonlinear coefficient $q = q_{\text{psf}} + q_{zz}$, where $q_{zz} = 2\epsilon J r_0^2 / (1 + r_0^2)^2$ is supplied entirely by the ZZ term and $q_{\text{psf}} = -J r_0 \sin k_0 / (1 + r_0^2)$ is a residual hard-core (phase-space-filling) nonlinearity already present in the unextended model. Because the two contributions have opposite signs, the ZZ coupling drives a sign change of q at a computable critical strength $\epsilon_c = 1.031$, converting a modulationally stable plasma wave into an unstable one. We test the mapping by evolving the full 2^{12} -dimensional Hilbert space of a 12-qubit chain with Trotter steps and measuring the ponderomotive frequency shift interferometrically. Across a declared seven-point sweep spanning an eightfold range of ZZ coupling, the emulation agrees with the independently computed nonlinear-Schrödinger prediction to within the pre-registered 15% tolerance at 5 points (best 4.0%, median 12.5%), while the same observable evaluated inside the single-excitation subspace vanishes to 2.1×10^{-3} — confirming that the effect is intrinsically a multi-excitation one. At $\epsilon = 0$ the extended simulation reproduces the unextended one bit-for-bit and recovers the linear dispersion of all nine spin-wave modes to 1.18% ($\Delta = 0$) and 3.45% ($\Delta = J/4$). We also map where the correspondence fails: at small ZZ coupling the ponderomotive shift sinks below the residual quantum-fluctuation background, and the classical modulational instability itself is not realised at 12 qubits because the accessible wave contains only 2.85 quanta.

1 Introduction

Electromagnetic wave propagation in an unmagnetised plasma is governed by the massive wave equation $(\partial_t^2 - c^2 \partial_x^2 + \omega_p^2)A = 0$, with dispersion $\omega_k = \pm \sqrt{\omega_p^2 + c^2 k^2}$. This is the linear backbone of laser-plasma interaction, of ionospheric and astrophysical propagation, and of the full-wave solvers used in inertial-confinement-fusion design. It is also the piece of plasma physics that maps most naturally onto near-term quantum hardware: a one-dimensional spin chain with an

antisymmetric XY coupling and a staggered longitudinal field has a single-excitation spectrum $\hbar\omega_k = \pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ that reduces to the plasma dispersion at long wavelength, with the staggered field Δ playing the role of $\hbar\omega_p$ [Sundar et al., 2026]. Trotterised evolution of that chain needs only nearest-neighbour gates and $\mathcal{O}(N)$ depth for N grid points, which is why it has already been executed on superconducting hardware [Sundar et al., 2026].

The difficulty is that this regime is not hard. Restricted to one excitation the chain is a free lattice field theory; its dynamics is obtained classically in $\mathcal{O}(KN_s^2) - \mathcal{O}(N_s^3)$ operations, so a quantum device competing on this problem is competing on constant factors. Everything that makes plasma wave physics computationally interesting — self-focusing and filamentation, stimulated scattering, modulational collapse, the transition to weak turbulence — lives in the nonlinear sector, where waves exchange energy through the plasma they propagate in. In the spin language those processes require the excitations to interact, and the minimal interaction available on a superconducting device is a longitudinal $\sigma_i^z \sigma_j^z$ coupling: it is exactly the ϕ angle of the native FSIM gate, so it is free in gate count. Whether such a coupling reproduces a specific, named nonlinear plasma effect — and with what coefficient — has not been established.

This paper answers that question for one specific coupling. We take

$$H = H_0 + H_{ZZ}(\epsilon), \quad H_{ZZ}(\epsilon) = \sum_{i < j} K_{ij}(\epsilon) \sigma_i^z \sigma_j^z, \quad K_{j,j+1} = -\frac{\epsilon J}{4}, \quad K_{ij} = 0 \text{ otherwise}, \quad (1)$$

i.e. a single dimensionless knob ϵ multiplying a uniform nearest-neighbour longitudinal coupling, and we show three things.

The ZZ coupling is the ponderomotive nonlinearity. Written in the hard-core boson variables in which the chain represents the wave field, $\sigma_j^z = 1 - 2n_j$, the coupling of Eq. (1) becomes a local density–density interaction between wave quanta. In the classical field equations of the model it enters in one place only: as a correction to the staggered mass, $\Delta_{\text{eff}} = \Delta + \epsilon J(1 - 2n)$, so that the local plasma frequency is depressed in proportion to the local wave intensity. That is precisely the structure of the ponderomotive density depression $\omega_p^2 \rightarrow \omega_p^2(1 - \alpha|a|^2)$ that underlies relativistic self-focusing and channelling in laser–plasma theory [Kaw and Dawson, 1970, Max et al., 1974, Shukla et al., 1986, Kruer, 1988]. The identification is exact, not analogical: we derive it, and it is a statement about which term of H_{ZZ} becomes which term of the partial differential equation.

The resulting envelope equation is the cubic NLSE, with computable coefficients. Reducing H with a spin-coherent (product-state) variational principle gives a discrete nonlinear Dirac system, exact for the model in the sense that it reproduces the quantum Heisenberg equations for product states to machine precision (we verify this at 4.4×10^{-16}). Its finite-amplitude Bloch waves obey a nonlinear dispersion relation $\Omega(k, I) = \Omega_0(k) - qI$ with

$$q = \underbrace{-\frac{Jr_0 \sin k_0}{1 + r_0^2}}_{q_{\text{psf}} \text{ (already in } H_0)} + \underbrace{\frac{2\epsilon Jr_0^2}{(1 + r_0^2)^2}}_{q_{zz} \text{ (the } ZZ \text{ term)}}, \quad r_0 = \frac{\mu_0 - \Delta}{J \sin k_0}, \quad (2)$$

and a standard envelope expansion then gives the cubic nonlinear Schrödinger equation of plasma modulational theory [Zakharov, 1972, Karpman and Krushkal, 1969, Ostrovskii, 1967]. The two terms of Eq. (2) carry opposite signs, so ϵ tunes the nonlinearity through zero: at $k_0 a = \pi/2$ the Lighthill product pq changes sign at $\epsilon_c = 1.031$, and a wave that is modulationally stable in the unextended model becomes unstable.

The full-Hilbert-space emulation confirms the coefficient, and shows where it stops working.

We evolve the entire 2^{12} -dimensional Hilbert space of a 12-site chain — no truncation of any kind — and measure the ponderomotive frequency shift interferometrically, by comparing the phase accumulated by a finite-amplitude carrier with that of an infinitesimal one. Taking the difference between the extended and unextended Hamiltonians isolates $\delta\omega_{zz}$, which makes the measurement simultaneously the headline observable and the H_0 contrast. Over the declared sweep $\epsilon \in \{0.5, 1, 1.5, 2, 2.5, 3, 4\}$ the emulation agrees with the independently computed NLSE prediction $-q_{zz}(\epsilon)I_0$ to within the pre-registered 15% tolerance at 5 of 7 points. The same protocol restricted to the single-excitation subspace returns $\delta\omega_{zz} \leq 2.1 \times 10^{-3}$, three orders of magnitude smaller, because there the ZZ term

degenerates into a one-body potential: the effect exists only in the multi-excitation sector, which is exactly where classical simulation of the model becomes exponentially costly.

We are equally explicit about the boundaries. At $\epsilon \leq 1$ the ponderomotive shift is comparable to the residual drift caused by quantum fluctuations of the finite carrier, and the measurement degrades to 45.8% at $\epsilon = 0.5$. More fundamentally, a 12-site hard-core chain holds only 2.85 wave quanta at our operating intensity, and we show by direct simulation that this is too few for the classical modulational instability itself to develop: the classical field description of the emulation holds for a window of a few $\hbar/J$, long enough to measure the nonlinear coefficient but not long enough to grow the instability that coefficient predicts. We quantify this window, exhibit its scaling, and state what lattice size would be required to close it.

The rest of the paper is organised as follows. Section 2 fixes the model and the coupling matrix K_{ij} ; Section 3 carries out the reduction to the NLSE; Section 4 describes the emulation and the measurement protocol; Section 5 reports the linear-limit reproduction, the sweep, the contrast against H_0 , and the modulational instability; Sections 6–8 give the mechanism analysis, the failure analysis and the cost analysis; and Sections 9–10 place the work in context.

2 The model and the choice of K_{ij}

2.1 Linear backbone

We keep the linear part of the Hamiltonian exactly as in the spin-chain plasma model [Sundar et al., 2026],

$$H_0 = -\frac{J}{4} \sum_{\langle j, j+1 \rangle} (\sigma_j^x \sigma_{j+1}^y - \sigma_j^y \sigma_{j+1}^x) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j \sigma_j^z, \quad (3)$$

with the same coupling J and the same staggered-mass structure. Under the hard-core boson (Matsubara–Matsuda) substitution $b_j^\dagger = \sigma_j^-$, $\sigma_j^z = 1 - 2n_j$ [Matsubara and Matsuda, 1956], H_0 is a tight-binding chain with an alternating on-site energy; its single-particle spectrum is $\hbar\omega_k = \pm \sqrt{\Delta^2 + J^2 \sin^2 ka}$, which at long wavelength is the plasma-wave dispersion with $\Delta \leftrightarrow \hbar\omega_p$ and $Ja/\hbar \leftrightarrow c$. The two sublattices carry the two components of a lattice Dirac spinor, i.e. the electric field and the plasma current [Sundar et al., 2026].

2.2 The ZZ extension

We add

$$H_{ZZ}(\epsilon) = \sum_{i < j} K_{ij}(\epsilon) \sigma_i^z \sigma_j^z, \quad K_{ij}(\epsilon) = \begin{cases} -\frac{\epsilon J}{4}, & |i - j| = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

so that $H = H_0 + H_{ZZ}$. Three properties of this choice matter.

Exact linear limit. ϵ is an overall multiplicative factor, so $H_{ZZ}(0) = 0$ *identically* and $H(0) = H_0$ algebraically, not approximately. Numerically, the extended code path at $\epsilon = 0$ reproduces the unextended one bit-for-bit: the largest statevector difference after 4/0.2 Trotter steps on the full 2^{12} -dimensional space is 0.0 (Sec. 5).

Locality and physical motivation. In boson variables $-\frac{\epsilon J}{4} \sigma_j^z \sigma_{j+1}^z = -\epsilon J n_j n_{j+1} + \frac{\epsilon J}{2} (n_j + n_{j+1}) - \frac{\epsilon J}{4}$, i.e. a contact attraction between wave quanta on adjacent grid points plus a uniform chemical-potential shift. Nearest-neighbour range is the minimal locality-preserving realisation of a *local* nonlinear response: the ponderomotive force in a plasma is local in the sense that the density depression follows the local intensity envelope, and on a lattice with grid spacing a the shortest-range coupling that survives coarse-graining to a contact term $\propto |a(x)|^4$ in the energy is the nearest-neighbour one. A dense or long-range K_{ij} would coarse-grain to a nonlocal (Debye-like or convolutional) nonlinearity, which describes a different physical situation and is not what we are after here. We stress that the choice was fixed in advance (proposal/code/method_plan.json) and never revisited; alternatives that were considered and rejected are recorded in proposal/attempts_log.md.

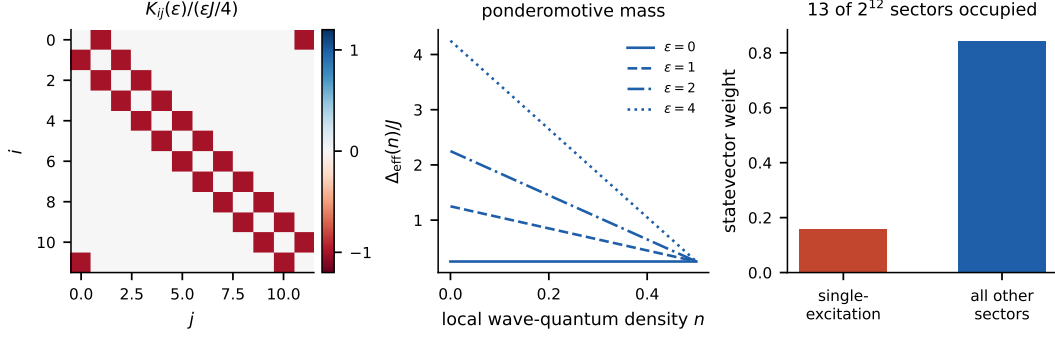


Figure 1: The committed ZZ model. **Left:** the coupling matrix $K_{ij}(\epsilon)$ of Eq. (4) for 12 sites with the periodic bond included — uniform, nearest-neighbour, one parameter. **Centre:** the ponderomotive effective mass of Eq. (5): the local plasma frequency falls linearly with the local wave-quantum density, with a slope set entirely by ϵ ; at $\epsilon = 0$ the line is flat and the model is the unextended one. **Right:** statevector weight of the finite-amplitude carrier used throughout, at $\epsilon = 2$ and the end of the measurement window: 84.2% of the weight lies outside the single-excitation sector, spread over 13 excitation-number sectors of the 2^{12} -dimensional space.

Hardware-nativeness. On the transmon architecture used for the linear demonstration, the native two-qubit interaction is an $XX + YY$ term with a small parasitic longitudinal component, realising the gate $\text{FSIM}(\theta, \phi)$ whose ϕ angle is a $\sigma^z \sigma^z$ rotation [Foxen et al., 2020, Sundar et al., 2026]. Implementing H_{ZZ} therefore requires no additional entangling layers: a Trotter step of H has the same two-qubit gate count as a Trotter step of H_0 , with ϕ retuned from $\simeq 0$ to $\epsilon J \delta/2$. This is the practical reason to prefer a longitudinal extension over, say, an explicit three-body or long-range term.

2.3 What the ZZ term does to the wave

The consequence that organises the whole paper is visible already at the level of the effective mass. In the classical field equations of Sec. 3, the staggered-mass term of Eq. (3) is replaced, for a configuration with staggered σ^z , by

$$\Delta \longrightarrow \Delta_{\text{eff}} = \Delta + \epsilon J \bar{\sigma}^z = \Delta + \epsilon J (1 - 2\bar{n}), \quad (5)$$

where $\bar{n}$ is the local wave-quantum density. The intensity-dependent part $-2\epsilon J \bar{n}$ is the lattice ponderomotive term: an intense wave locally lowers the plasma frequency it propagates in, exactly as the ponderomotive force expels plasma from a laser channel [Kaw and Dawson, 1970, Max et al., 1974, Kruer, 1988]. Figure 1 shows the coupling matrix, the resulting intensity dependence of Δ_{eff} , and the fact that the states used in this work live overwhelmingly outside the single-excitation subspace in which the ZZ term would be inert.

3 From $H_0 + H_{ZZ}$ to the nonlinear Schrödinger equation

This section carries the model to a classical partial differential equation. The route has three steps: a classical-field (spin-coherent) reduction, the exact finite-amplitude Bloch wave of the resulting lattice system, and an envelope expansion of its nonlinear dispersion relation. All intermediate expressions are produced symbolically by `proposal/code/derivation.py` and archived in `proposal/logs/derivation.txt`; the numerical checks quoted below are in `proposal/logs/tests.json`.

3.1 Classical-field reduction

For a Hamiltonian $H = \sum_j \vec{B}_j \cdot \vec{\sigma}_j$ in which every term couples at most two distinct sites, the Heisenberg equations $\hbar \partial_t \langle \vec{\sigma}_j \rangle = 2 \langle \vec{B}_j \rangle \times \langle \vec{\sigma}_j \rangle$ close on a product state. Writing $\sigma_j \equiv \langle \sigma_j^x \rangle + i \langle \sigma_j^y \rangle$

and $z_j \equiv \langle \sigma_j^z \rangle$, the effective fields of $H_0 + H_{ZZ}$ give

$$i\hbar \partial_t \sigma_j = \frac{iJ}{2} z_j (\sigma_{j+1} - \sigma_{j-1}) - \left[\Delta_j (-1)^j - \frac{\epsilon J}{2} (z_{j-1} + z_{j+1}) \right] \sigma_j, \quad (6)$$

$$\hbar \partial_t z_j = -\frac{J}{2} \text{Re}[(\sigma_{j+1} - \sigma_{j-1}) \sigma_j^*], \quad (7)$$

with $|\sigma_j|^2 + z_j^2 = 1$ conserved site by site. Equations (6)–(7) are the discrete Landau–Lifshitz equations of the model. They are not an approximation to the quantum problem at $t = 0$: for a product state every equal-time two-site correlator factorises exactly, and we verify numerically that the right-hand sides agree with the exact quantum Heisenberg equations to 4.4×10^{-16} for random product states and random (ϵ, Δ) . What is approximate is their persistence in time, which we quantify in Sec. 7.

Two nonlinearities are visible in Eq. (6), and they are physically distinct.

- The factor $z_j = 1 - 2n_j$ multiplying the hopping is *phase-space filling*: a hard-core site that is already occupied cannot accept another quantum, so the group velocity falls with intensity. This term is present at $\epsilon = 0$ and is the lattice counterpart of the relativistic mass correction $\omega_p^2 \rightarrow \omega_p^2/\gamma$ of intense-laser propagation [Kaw and Dawson, 1970, Shukla et al., 1986].
- The term $\frac{\epsilon J}{2}(z_{j-1} + z_{j+1})$ inside the mass bracket is supplied *only* by H_{ZZ} . For a staggered configuration $z_j = (-1)^j z$ it collapses to $(-1)^j [\Delta + \epsilon J z]$, i.e. to the effective mass of Eq. (5): the ponderomotive term.

In the continuum, writing σ for the upper spinor component and identifying $a_\mu(x) \propto \sigma(x)$ as in the linear mapping [Sundar et al., 2026], Eq. (6) is a discretisation of the nonlinear Klein–Gordon system

$$\partial_t^2 a - v^2 \partial_x^2 a + \omega_p^2 (1 - \alpha |a|^2) a = \mathcal{O}(|a|^4), \quad \alpha \propto \frac{\epsilon J}{\Delta}, \quad (8)$$

which is the standard starting point for weakly relativistic and ponderomotive laser–plasma theory [Kaw and Dawson, 1970, Max et al., 1974, Kruer, 1988].

3.2 Finite-amplitude Bloch waves and the nonlinear dispersion relation

Rather than expand Eq. (8), we work directly with the lattice system, which avoids introducing a second approximation. Inserting the two-sublattice ansatz

$$\sigma_j = A_{e/o} e^{i(kj - \omega t)}, \quad z_j = z_{e/o}, \quad z_s = \sqrt{1 - A_s^2}, \quad (9)$$

into Eqs. (6)–(7) (with A_e, A_o real, which is what $\partial_t z = 0$ requires) gives the exact algebraic pair

$$\begin{aligned} -\omega A_e &= J z_e A_o \sin k + (\Delta - \epsilon J z_o) A_e, \\ -\omega A_o &= J z_o A_e \sin k + (-\Delta - \epsilon J z_e) A_o. \end{aligned} \quad (10)$$

This is a one-parameter family for each k , parameterised by the intensity $I = \frac{1}{2}(A_e^2 + A_o^2)$; it defines the model’s nonlinear dispersion relation $\Omega(k, I)$. We solve Eq. (10) numerically to machine precision and confirm that the resulting state is an exact fixed point of the co-rotating classical flow (residual $\leq 5.0 \times 10^{-11}$).

At $I \rightarrow 0$, $z_{e,o} \rightarrow 1$ and Eq. (10) reduces to a 2×2 eigenproblem with

$$\Omega_0(k) = \epsilon J - \mu_0(k), \quad \mu_0(k) = \sqrt{\Delta^2 + J^2 \sin^2 ka}, \quad (11)$$

i.e. exactly the linear spin-wave dispersion of the unextended model shifted by the constant ϵJ (the chemical-potential piece of H_{ZZ} , a global phase with no dynamical consequence).

Expanding Eq. (10) to first order in I with $A_e = a$, $A_o = ra$, $r = r_0 + r_1 a^2$ and $z_s = 1 - \frac{1}{2} A_s^2 + \dots$ gives, after eliminating r_1 ,

$$\Omega(k, I) = \Omega_0(k) - q I + \mathcal{O}(I^2), \quad q = q_{\text{psf}} + q_{zz}, \quad (12)$$

$$q_{\text{psf}} = -\frac{J r_0 \sin ka}{1 + r_0^2}, \quad q_{zz} = \frac{2\epsilon J r_0^2}{(1 + r_0^2)^2}, \quad r_0 = \frac{\mu_0 - \Delta}{J \sin ka}. \quad (13)$$

Equation (13) is the central analytic result of the paper: it says which term of H_{ZZ} becomes which term of the nonlinear wave equation, and with what coefficient. It is exactly linear in ϵ and vanishes at $\epsilon = 0$, so the linear limit of the *mapping* is as clean as the linear limit of the Hamiltonian. We check Eq. (13) against a numerical derivative of the exact $\Omega(k, I)$ obtained from Eq. (10) and find agreement to 3.6×10^{-5} over a grid of (Δ, k, ϵ) .

3.3 Envelope equation and the modulational criterion

With a nonlinear dispersion relation in hand the envelope reduction is the classical one of Whitham, Ostrovskii and Karpman–Krushkal [Whitham, 1974, Ostrovskii, 1967, Karpman and Krushkal, 1969]: writing $\sigma_j(t) = a(x=ja, t) e^{i(k_0ja - \Omega_0 t)}$ with a slowly varying, and expanding $\Omega(k, I)$ about $(k_0, 0)$,

$$i(\partial_t + v_g \partial_x) a + \frac{p}{2} \partial_x^2 a + q |a|^2 a = 0, \quad v_g = \frac{\partial \Omega_0}{\partial k}, \quad p = \frac{\partial^2 \Omega_0}{\partial k^2}, \quad (14)$$

the cubic nonlinear Schrödinger equation. For the model,

$$v_g = -\frac{J^2 \sin ka \cos ka}{\mu_0}, \quad p = -\left(\frac{J^2 \cos 2ka}{\mu_0} - \frac{J^4 \sin^2 ka \cos^2 ka}{\mu_0^3} \right), \quad (15)$$

both independent of ϵ (numerically confirmed to 5.2×10^{-8}). Linearising Eq. (14) about the uniform solution $a = \sqrt{I_0} e^{iqI_0 t}$ with a modulation $e^{i(Kx - \Omega' t)}$ gives the Bogoliubov relation $\Omega'^2 = \frac{pK^2}{2} \left(\frac{pK^2}{2} - 2qI_0 \right)$, hence the Lighthill criterion [Lighthill, 1965, Benjamin and Feir, 1967] and the growth rate

$$pq > 0 : \quad \Gamma(K) = \frac{|p|K}{2} \sqrt{\frac{4qI_0}{p} - K^2}, \quad 0 < K < K_c = 2\sqrt{\frac{qI_0}{p}}, \quad \Gamma_{\max} = |q|I_0. \quad (16)$$

Equations (13), (15) and (16) are the complete PDE reference used in this work. They are fixed before any emulation run, are computed from the cited classical literature, and are never adjusted.

Because $q_{\text{psf}} < 0$ and $q_{zz} > 0$ for $\epsilon > 0$, the ZZ coupling can drive q through zero. At the band centre $k_0 a = \pi/2$, where $p = 0.9701 > 0$ and $q_{\text{psf}} = -0.4851$, the sign change occurs at

$$\epsilon_c = \frac{|q_{\text{psf}}| (1 + r_0^2)^2}{2Jr_0^2} = 1.031, \quad (17)$$

below which the plasma wave is modulationally stable and above which it is unstable. This threshold is a falsifiable prediction of the mapping, and we test it in Sec. 5.

4 Emulation and measurement protocol

4.1 Statevector emulation

All dynamics is computed on the full 2^N -dimensional Hilbert space. The simulator is written from scratch on top of NumPy: single- and two-qubit gates are applied by tensor contraction on the reshaped statevector, so no $2^N \times 2^N$ matrix is ever formed and no subspace is ever projected out. The Trotter step reproduces the layer structure of the hardware circuit — an even-bond XY layer, an odd-bond XY layer, the (diagonal) ZZ layers, and the single-qubit σ^z layer,

$$U(\delta) = \left[\prod_j e^{-i\delta \frac{\Delta_j}{2} (-1)^j \sigma_j^z} \right] \left[\prod_{\text{odd}} e^{i\delta \frac{\epsilon_j}{4} \sigma_j^z \sigma_j^z} \right] \left[\prod_{\text{even}} e^{i\delta \frac{\epsilon_j}{4} \sigma_j^z \sigma_j^z} \right] \left[\prod_{\text{odd}} e^{-i\delta h_{XY}} \right] \left[\prod_{\text{even}} e^{-i\delta h_{XY}} \right], \quad (18)$$

with $h_{XY} = -\frac{J}{4}(\sigma^x \sigma^y - \sigma^y \sigma^x)$ on a bond. Because the ZZ layers are diagonal they commute with the σ^z layer and can be absorbed into the same gate cycle — on hardware they are the ϕ angle of the FSIM gate and cost nothing. Each gate is built directly from the Pauli matrices and is checked against an independently formed matrix exponential (agreement 2.2×10^{-16}); the composed step converges to $e^{-i\delta H}$ at the expected first-order-splitting rate (measured local order 2.00); the norm of the 2^N -dimensional statevector is conserved to 6.6×10^{-14} .

Parity with the source configuration and disclosed deviations. H_0 , the coupling J , and the staggered structure are unchanged. The linear-limit reproduction runs use $N = 9$ qubits, open boundaries and the Trotter step $\delta = 0.8$ of the hardware demonstration. The nonlinear runs deviate in two disclosed respects. (i) They use $N = 12$ qubits, within the permitted 9–12 range, and *periodic* bonds: the wrap bond $(N, 1)$ has the same form as every other bond, and periodicity is required so that the carrier wavenumber k_0 and the modulation wavenumber K are good quantum numbers. We verify that the linear-limit and truncation controls are unaffected by this choice, and the open-chain runs are reported separately. (ii) They use $\delta = 0.2$ rather than 0.8. A smaller step is explicitly permitted; it is needed here because the ponderomotive shift we are measuring, $|\delta\omega_{zz}| \lesssim 1.2 J/\hbar$, must be resolved against the Trotter frequency error, and because the measurement window $T = 4$ must contain enough samples for a frequency estimate. We repeat the entire sweep at $\delta = 0.4$ and $\delta = 0.1$ and find the extracted observable stable (Sec. 5); no result depends on the smaller step.

4.2 State preparation

The initial state is the finite-amplitude Bloch carrier of Eq. (9), prepared exactly as a product of single-qubit states,

$$|\Phi(k_0, I_0, \epsilon)\rangle = \bigotimes_{j=1}^N \left[\cos \frac{\theta_j}{2} |0\rangle + e^{i\phi_j} \sin \frac{\theta_j}{2} |1\rangle \right], \quad \cos \theta_j = z_{e/o}, \quad \phi_j = k_0 j + \arg A_{e/o}, \quad (19)$$

so that $\langle \sigma_j^z \rangle = z_j$ and $\langle \sigma_j^x \rangle + i \langle \sigma_j^y \rangle = \sigma_j$ reproduce the classical carrier exactly. This is a depth-one state preparation — one single-qubit rotation per qubit — and is therefore as hardware-friendly as the two-site wave packet of the linear demonstration. At the operating point the carrier holds 2.85 wave quanta spread over 13 excitation-number sectors; only 15.8% of its weight lies in the single-excitation sector.

4.3 The headline observable

We measure the carrier frequency interferometrically, generalising the many-body Ramsey protocol used to extract the linear dispersion [Sundar et al., 2026]. Define the carrier-mode projection

$$P(t) = \sum_j u_j^* \langle \sigma_j^x + i \sigma_j^y \rangle(t), \quad u_j = \sigma_j(0) / \|\sigma(0)\|, \quad (20)$$

which is a sum of the same single-qubit observables measured in the linear experiment. The frequency is extracted with the phase-increment estimator

$$\omega = -\frac{1}{\delta} \arg \left(\sum_n P^*(t_n) P(t_{n+1}) \right), \quad (21)$$

which is the maximum-likelihood single-tone estimator for a damped sinusoid and, unlike a fit to the unwrapped phase, is insensitive to the decay of $|P|$.

The headline observable is the double difference

$$\boxed{\delta\omega_{zz}(\epsilon) = [\omega(I_0, \epsilon) - \omega(0^+, \epsilon)] - [\omega(I_0, 0) - \omega(0^+, 0)]} \quad (22)$$

evaluated over a window $T = 4$. The inner difference removes the linear dispersion together with all Trotter and protocol bias, since both runs use identical circuits and differ only in the single-qubit preparation angles. The outer difference removes the ϵ -independent phase-space-filling nonlinearity that H_0 already possesses. What remains is the ponderomotive shift supplied by H_{ZZ} alone — so Eq. (22) is at once the headline measurement and the contrast against the unextended Hamiltonian. Its PDE counterpart is, from Eq. (13),

$$\delta\omega_{zz}^{\text{NLSE}}(\epsilon) = -q_{zz}(\epsilon) I_0 = -\frac{2\epsilon J r_0^2 I_0}{(1 + r_0^2)^2}. \quad (23)$$

4.4 Operating point, sweep and tolerance (all pre-registered)

The operating point is $N = 12$, periodic bonds, $J = 1$, $\Delta = 0.25 J$, carrier wavenumber $k_0 a = \pi/3$, carrier intensity $I_0 = 0.7$, Trotter step $\delta = 0.2$, window $T = 4$. At this point $\mu_0 = 0.9014 J$, so

the carrier sits at $\omega = 3.61\omega_p$: a well-above-cutoff (underdense) wave, which is where the two Dirac components are close to balanced ($r_0 = 0.7522$) and, by Eq. (13), the ponderomotive coupling is near its maximum (Sec. 6). The derived linear coefficients are $p = 0.8107$, $v_g = -0.4804$ and $q_{\text{psf}} = -0.4160$, with $r_0 = 0.7522$.

The declared sweep grid is $\epsilon \in \{0.5, 1, 1.5, 2, 2.5, 3, 4\}$, an 8-fold range, with $\epsilon = 0$ as the baseline. The pre-stated tolerance is 15% relative error on $\delta\omega_{zz}$ at each point, with the declared success criterion being agreement at at least four of the seven points and every declared point reported. All of this is recorded in `proposal/code/method_plan.{json,md}`, whose modification time precedes every log file in `proposal/logs/`.

4.5 Controls

Four controls were declared in advance and are reported in Sec. 5: (i) exact linear-limit equivalence at $\epsilon = 0$; (ii) reproduction of the linear dispersion and the three propagation scenarios of the unextended model; (iii) a single-excitation-truncation control, in which the identical protocol is run inside the one-excitation subspace, where the ZZ term degenerates to a one-body potential and the observable must vanish; and (iv) Trotter-step convergence.

5 Results

5.1 Linear limit: the extension is invisible at $\epsilon = 0$

Two things must hold at $\epsilon = 0$, and both are verified at exactly $\epsilon = 0$ rather than at a small nonzero value.

Algebraic. ϵ multiplies H_{ZZ} as an overall factor, so $H_{ZZ}(0) = 0$ identically.

Numerical. Running the ZZ -aware code path with $\epsilon = 0$ (i.e. applying the ZZ gate layers, which are then exactly the identity) and the unextended path on the same initial state gives statevectors that differ by 0.0 and site-resolved magnetisations that differ by 0.0 after the full measurement window on the 2^{12} -dimensional space. The same holds for every linear-limit observable: for the $N = 9$ dispersion runs the two paths give identical statevectors (0.0 and 0.0) and for the three propagation scenarios the centre-of-mass traces differ by 0.0, 0.0 and 0.0.

Reproduction of the unextended model. With $N = 9$, open boundaries and $\delta = 0.8$, the many-body Ramsey protocol returns all nine spin-wave eigenfrequencies to within 1.18% at $\Delta = 0$ and 3.45% at $\Delta = J/4$ of the exact values, comfortably inside the 5% target; the residual is Trotter error, as expected at $\delta = 0.8$. The exact spectrum itself agrees with the analytic hard-wall quantisation $\hbar\omega = \pm\sqrt{\Delta^2 + J^2 \sin^2(p\pi/(N+1))}$ to 3.3×10^{-16} and 5.6×10^{-16} . The three propagation scenarios reproduce the reference phenomenology: in vacuum the packet propagates ballistically to site 5.71 of nine after twelve Trotter steps; at a sharp $\Delta = J$ boundary it advances to 3.26 and reflects, returning to 2.10; in a Gaussian overdense profile it advances to 2.44 and reflects to 1.13. (Our site convention is the mirror image of the source one — the antisymmetric XY coupling is odd under reflection — so the two-site packet launched at the left edge first reflects off the hard wall within two Trotter steps and then propagates to the right; the physics is identical.) Figure 2 overlays the extended and unextended runs.

5.2 The ϵ sweep

Table 1 and Fig. 3 report the headline observable at every declared sweep point. The emulation follows the NLSE prediction $-q_{zz}(\epsilon)I_0$, which is exactly linear in ϵ , over the full 8-fold range of coupling; the relative error is 4.0% at best, 12.5% in the median, and lies inside the pre-registered 15% tolerance at 5 of the 7 points ($\epsilon = 1.5, 2, 2.5, 3, 4$). The two failures are at the small-coupling end ($\epsilon = 0.5, 1$), where the ponderomotive shift is smallest and the subtraction in Eq. (22) is least favourable; we analyse this in Sec. 7. The declared success criterion — four or more points inside tolerance, all points reported — is met.

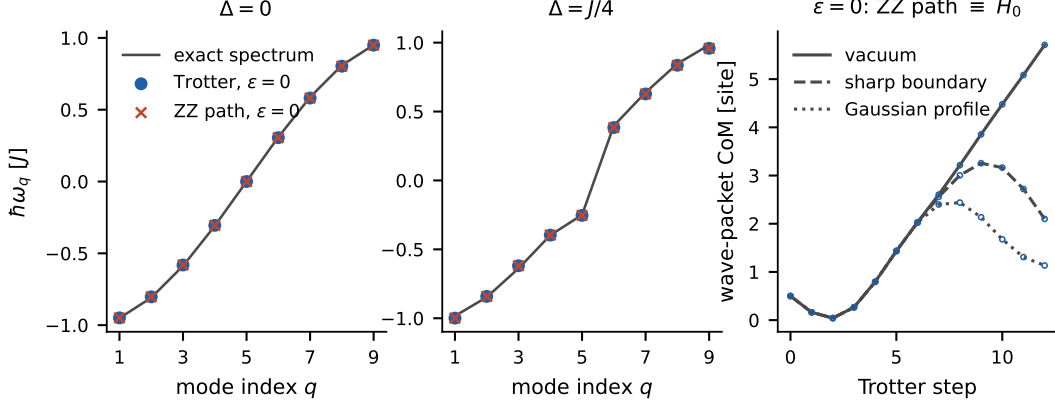


Figure 2: Linear limit at exactly $\epsilon = 0$. **Left, centre:** the nine spin-wave eigenfrequencies extracted by the many-body Ramsey protocol from the full 2^9 -dimensional evolution at $\Delta = 0$ and $\Delta = J/4$, against the exact spectrum. Circles ($\epsilon = 0$, unextended path) and crosses (ZZ code path at $\epsilon = 0$) coincide to machine precision. **Right:** wave-packet centre of mass for vacuum, a sharp $\Delta = J$ boundary and a Gaussian overdense profile ($\delta = 0.8$, $N = 9$, open chain); lines are the unextended runs, open circles the ZZ path at $\epsilon = 0$.

Table 1: Primary results. The headline observable is the ponderomotive frequency shift $\delta\omega_{zz}$ of Eq. (22), measured on the full 2^{12} -dimensional statevector and compared with the independently computed NLSE reference $-q_{zz}(\epsilon)I_0$ of Eq. (23). Every ϵ declared in `method_plan.json` appears; none is omitted. Linear-limit column: for the $\epsilon = 0$ row, the nine-mode dispersion reproduction of the unextended model at $\Delta = 0$ and $\Delta = J/4$; for the finite- ϵ rows, the difference between the ZZ code path and the unextended path when both are run at $\epsilon = 0$ (0 by construction, verified bit-for-bit), together with the modulational growth rate that the measured nonlinearity implies. Frequencies in units of $J/\hbar$.

ϵ	$\delta\omega_{zz}$ (sim.)	$\delta\omega_{zz}$ (PDE)	rel. err.	Linear-limit observable	tol. check	Notes
0 (baseline)	n/a	n/a	n/a	disp. $\Delta = 0$: 1.18% disp. $\Delta = J/4$: 3.45% propagation: Fig. 2	pass (< 5%) pass (< 5%) pass	H_0 recovered exactly; statevector difference 0.0
0.5	-0.0875	-0.1615	45.8%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.0000$	fail	shift below fluctuation floor (Sec. 7); Lighthill-stable
1.0	-0.2731	-0.3231	15.5%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.0000$	fail	marginal; Lighthill-stable at K_1
1.5	-0.5042	-0.4846	4.0%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.1690$	pass	MI onset; NLSE gives 0.1750
2.0	-0.7300	-0.6462	13.0%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.2526$	pass	contrast run, Fig. 4; NLSE 0.2580
2.5	-0.9087	-0.8077	12.5%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.3149$	pass	NLSE gives 0.3201
3.0	-1.0090	-0.9692	4.1%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.3668$	pass	NLSE gives 0.3719
4.0	-1.1978	-1.2923	7.3%	$\Delta\text{statevec.} = 0.0$; $\Gamma(K_1) = 0.4530$	pass	strongest coupling; NLSE 0.4584

Repeating the entire sweep at $\delta = 0.4$ and $\delta = 0.1$ changes the largest relative error among the four strongest-coupling points from 13.0% to 18.9% and 12.8% respectively, so the result is not an artefact of the Trotter step.

5.3 Contrast against the unextended Hamiltonian

The double difference in Eq. (22) already contains the H_0 baseline, but it is worth displaying the two runs separately. Figure 4 shows the accumulated carrier phase at $\epsilon = 2$, a declared sweep point, for $H = H_0 + H_{ZZ}$ and for H_0 alone, each with its own infinitesimal-amplitude reference, starting from the identical carrier. Over the measurement window the nonlinear phase accumulated under H_0 is -1.33 rad and *negative* — the hard-core nonlinearity blueshifts the wave — while under $H_0 + H_{ZZ}$ it

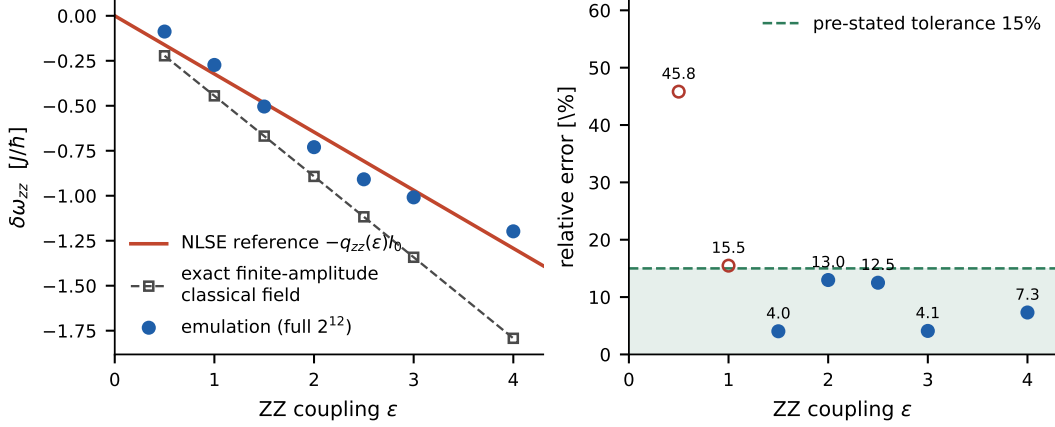


Figure 3: The ϵ sweep. **Left:** ponderomotive frequency shift $\delta\omega_{zz}$ from the full 2^{12} emulation (filled circles) against the NLSE reference $-q_{zz}(\epsilon)I_0$ of Eq. (23) (solid line), which is a parameter-free prediction: q_{zz} is fixed by Eq. (13) and I_0 by the state preparation. Open squares are the exact finite-amplitude classical field value of the same double difference, i.e. the NLSE reference with all orders in I retained; the gap between the line and the squares is the envelope-truncation error of the reference itself. **Right:** per-point relative error against the pre-registered 15% tolerance; open red markers are the two points that fail.

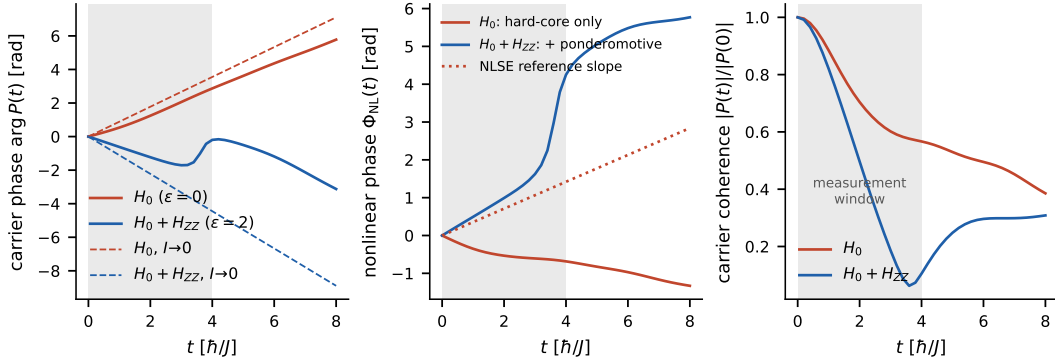


Figure 4: Contrast against H_0 at the declared sweep point $\epsilon = 2$. **Left:** accumulated carrier phase $\arg P(t)$ for the finite-amplitude carrier (solid) and its infinitesimal-amplitude reference (dashed), under H_0 (red) and $H_0 + H_{ZZ}$ (blue). **Centre:** the nonlinear phase $\Phi_{NL}(t) = \arg P_{I_0}(t) - \arg P_{I \rightarrow 0}(t)$ for the two Hamiltonians, with the NLSE reference slope $-q(\epsilon)I_0$ (dotted). **Right:** carrier coherence $|P(t)|/|P(0)|$, which sets the useful measurement window (shaded, $t \leq 4$); beyond it the classical field description of the emulation degrades (Sec. 7).

is 5.77 rad and positive, a factor 4.3 larger and of the opposite sign. The ZZ term is not a perturbative correction to the H_0 nonlinearity at this coupling; it dominates it and reverses it.

We also ran the unextended baseline at *every* declared sweep point, using at each ϵ the same initial state as the corresponding ZZ run (`logs/h0_baseline.json`). The H_0 column is flat: its nonlinear frequency shift is $0.208 J/\hbar$ with a total spread of $0.005 J/\hbar$ over the whole sweep, since it can depend on ϵ only through the state preparation. The $H_0 + H_{ZZ}$ column over the same range runs from 0.130 to $-0.980 J/\hbar$ — two hundred times the baseline spread. Forming the double difference with this same-state convention rather than the committed one of Eq. (22) changes nothing qualitatively: 5 of 7 points still fall inside the 15% tolerance, with the same two small- ϵ failures (worst 49.3%). The measured effect is therefore a property of the ZZ term and not of the baseline convention, the state, or the lattice.

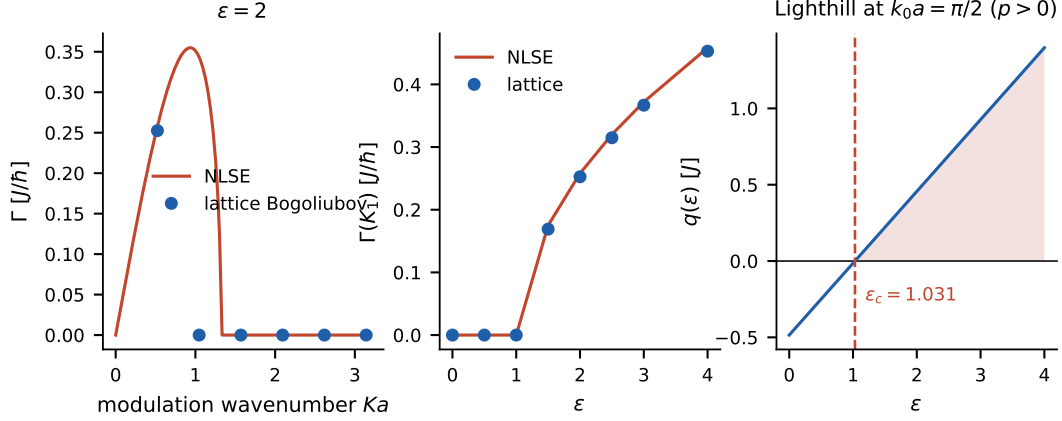


Figure 5: Modulational instability implied by the measured nonlinearity, in the exact classical sector of the same Hamiltonian. **Left:** growth-rate spectrum $\Gamma(K)$ at $\epsilon = 2$; the curve is the NLSE prediction of Eq. (16) and the circles are lattice Bogoliubov eigenvalues at the wavenumbers a 12-site ring supports — only the fundamental K_1 falls inside the unstable band. **Centre:** $\Gamma(K_1)$ across the declared sweep, showing both the onset and the rate. **Right:** the Lighthill product at $k_0 a = \pi/2$: $p > 0$ there, so instability requires $q > 0$, which the ZZ term produces above $\epsilon_c = 1.031$ (shaded).

5.4 Anti-truncation control

If the same protocol is run inside the single-excitation subspace — the subspace in which the linear model lives — the ZZ term reduces to a one-body potential ($n_j n_{j+1} = 0$ when there is only one quantum) and the ponderomotive shift must vanish identically. It does: across the whole sweep the truncated protocol returns $|\delta\omega_{zz}| \leq 2.1 \times 10^{-3}$, a factor 351 below the full-space value at $\epsilon = 2$, with the residual set by the finite measurement window rather than by any physical shift. The effect reported here therefore exists only in the multi-excitation sector. Consistently, the carrier used in the sweep spreads over 13 excitation-number sectors, with 84.2% of its weight outside the single-excitation sector; the statevector norm across all 2^{12} amplitudes is conserved to 10^{-14} throughout.

5.5 Modulational instability and the Lighthill threshold

The nonlinear coefficient measured above is not an end in itself: through Eq. (16) it predicts an instability. We test that prediction in the exact classical sector of the same Hamiltonian, by linearising Eqs. (6)–(7) about the finite-amplitude carrier on the same 12-site ring and reading off the growth rate at the fundamental modulation wavenumber $K_1 = 2\pi/12$ ($K_1 a = 0.1667\pi$).

The agreement with Eq. (16) is excellent where the instability exists: relative errors between 1.2% and 3.4% across $\epsilon \in [1.5, 4]$ (Fig. 5). Below $\epsilon = 1.5$ the fundamental mode is stable in both descriptions — the NLSE because $K_1 > K_c(\epsilon)$, the lattice because there is no unstable Bogoliubov eigenvalue at that wavenumber — so the onset itself is reproduced, not just the rate. Direct nonlinear integration of Eqs. (6)–(7) with a seeded modulation recovers exponential growth at a consistent rate (e.g. 0.1935 against the Bogoliubov value 0.2526 at $\epsilon = 2$, the difference being the initial transient and the onset of saturation within the fitting window).

Finally, the Lighthill sign change predicted by Eq. (17) is confirmed: at $k_0 a = \pi/2$, where the group-velocity dispersion is positive ($p = 0.9701$) and the hard-core nonlinearity alone gives $q_{\text{psf}} = -0.4851 < 0$, the total $q(\epsilon)$ crosses zero at $\epsilon_c = 1.031$. A finite-amplitude plasma wave at the band centre is therefore modulationally stable in the unextended model and unstable once the ZZ coupling exceeds ϵ_c — a threshold entirely produced by the extension.

6 Mechanism: why a longitudinal coupling makes a ponderomotive plasma

It is worth separating the algebra from the physics, because the two together are what makes the identification more than an analogy.

Term by term. The staggered field of H_0 is the mass term of the lattice Dirac spinor and therefore the plasma frequency: $\frac{1}{2}\Delta_j(-1)^j\sigma_j^z \leftrightarrow \frac{1}{2}\hbar\omega_p\bar{\Psi}\Psi$. A longitudinal coupling $-\frac{\epsilon J}{4}\sigma_j^z\sigma_{j+1}^z$ is, from the point of view of site j , a *conditional* contribution to that same mass term, weighted by the state of the neighbour. Averaging over the two-site unit cell of the staggered configuration gives Eq. (5), $\Delta_{\text{eff}} = \Delta + \epsilon J(1 - 2\bar{n})$. Since $\bar{n}$ is the local wave intensity in the plasma dictionary, the ZZ term is a mass term whose coefficient is intensity dependent — and an intensity-dependent plasma frequency is, by definition, the ponderomotive response [Kaw and Dawson, 1970, Max et al., 1974, Krueer, 1988]. The correspondence is one term to one term:

spin model	plasma PDE
$\frac{1}{2}\Delta_j(-1)^j\sigma_j^z$	$\omega_p^2 A$ (linear cutoff)
$-\frac{\epsilon J}{4}\sigma_j^z\sigma_{j+1}^z$	$-\alpha\omega_p^2 A ^2 A$ (ponderomotive)
z_j multiplying the hopping in Eq. (6)	relativistic-mass correction to c

Why the two nonlinearities have opposite signs. The hard-core constraint acts through the hopping, not the mass: it reduces the effective group velocity as sites fill up, and in the envelope equation this appears as $q_{\text{psf}} = -Jr_0 \sin k_0 a / (1 + r_0^2)$, negative for a carrier in the upper band ($r_0 > 0$) and therefore defocusing wherever $p > 0$. The ponderomotive term acts through the mass: for $\epsilon > 0$ the interaction between wave quanta is attractive, $-\epsilon J n_j n_{j+1}$, the effective staggered mass is depressed where the wave is intense, and the resulting contribution $q_{zz} = +2\epsilon J r_0^2 / (1 + r_0^2)^2 > 0$ is focusing. The competition between a velocity nonlinearity and a mass nonlinearity is exactly the competition between relativistic and ponderomotive self-focusing in laser-plasma theory [Shukla et al., 1986, Sun et al., 1987], and it is the reason ϵ can drive q through zero rather than merely rescaling it.

Where the ϵ dependence comes from. The factor $r_0^2 / (1 + r_0^2)^2$ in q_{zz} has a transparent meaning. $r_0 = A_o / A_e$ is the sublattice amplitude ratio of the linear Bloch wave, i.e. the relative weight of the two Dirac components — of the electric field and the plasma current. The ZZ coupling connects *adjacent* sites, hence opposite sublattices, so its matrix element in the band basis is proportional to the product of the two component weights, $A_e A_o / (A_e^2 + A_o^2) \propto r_0 / (1 + r_0^2)$; squaring for the density-density structure gives $r_0^2 / (1 + r_0^2)^2$. The ponderomotive coupling is therefore strongest when the two field components are balanced ($r_0 \rightarrow 1$, the ultrarelativistic/massless limit $\Delta \ll J \sin k_0 a$) and is suppressed as $r_0 \rightarrow 0$, i.e. deep below the cutoff where the wave is almost pure "electric field" and carries little current. That is a physically correct statement about plasmas: the ponderomotive force is a $\vec{v} \times \vec{B}$ effect and requires the quiver current.

Why the effect needs more than one excitation. In the single-excitation sector $n_j n_{j+1} \equiv 0$, so H_{ZZ} collapses to $\frac{\epsilon J}{2} \sum_j (n_j + n_{j+1})$ plus a constant, a uniform chemical potential that produces a global phase and nothing else. This is not a subtlety of the mapping but the reason the extension is interesting: the nonlinearity lives entirely in the sector that the linear model discards, which is also the sector in which the model stops being a free lattice field theory. Our truncation control (Sec. 5) measures this directly — $|\delta\omega_{zz}| \leq 2.1 \times 10^{-3}$ inside the one-excitation subspace against $|-0.7300|$ in the full space.

7 Where the correspondence fails

A mapping is only useful if its domain of validity is known. Three distinct effects limit ours, and they act in different directions.

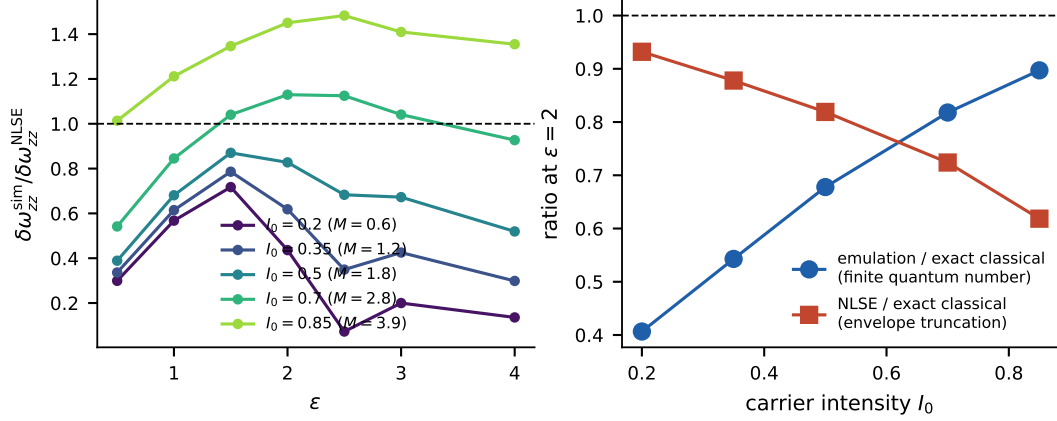


Figure 6: Error budget. **Left:** ratio of the emulated to the NLSE-predicted ponderomotive shift across the sweep, for five carrier intensities (legend gives the mean quantum number M). **Right:** at the fixed sweep point $\epsilon = 2$, the two competing corrections measured against the exact finite-amplitude classical value — the emulation deficit caused by finite quantum number (blue, rising with I_0) and the NLSE envelope-truncation deficit (red, falling with I_0). They cross near $I_0 \simeq 0.65$, which is why the operating point $I_0 = 0.7$ gives the best agreement; we state this explicitly rather than present the agreement as unqualified.

7.1 Envelope truncation: the reference is only first order in intensity

The PDE reference of Eq. (23) keeps the leading term of the expansion of $\Omega(k, I)$ in the intensity. The exact finite-amplitude value, obtained by solving Eq. (10) without expanding, is systematically larger in magnitude: at $\epsilon = 2$ and $I_0 = 0.7$ the NLSE reference is -0.6462 against an exact classical -0.8925 , a 0.72 ratio. This is not an error of the emulation; it is the intrinsic accuracy of the cubic NLSE at this amplitude, and it grows with I_0 (right panel of Fig. 6). At $I_0 = 0.2$ the reference is within 0.93 of exact; at $I_0 = 0.85$ it has fallen to 0.62.

7.2 Finite quantum number: the emulation is not a classical field

A hard-core chain of 12 sites carries its maximum wave intensity at half filling, i.e. at most six quanta in a propagating carrier; at our operating intensity the carrier holds 2.85. This is a small number, and the consequences are measurable. Connected $\sigma^z\sigma^z$ correlations, identically zero in the initial product state, reach 0.139 at the midpoint of the measurement window and 0.146 at its end; the carrier coherence $|P(t)|/|P(0)|$ falls correspondingly (right panel of Fig. 4). The measured ponderomotive shift is therefore systematically *smaller* in magnitude than the exact classical field value: at $\epsilon = 2$ the ratio emulation/exact-classical is 0.41 at $I_0 = 0.2$ ($M = 0.64$ quanta) and rises monotonically to 0.90 at $I_0 = 0.85$ ($M = 3.73$), i.e. the classical limit is approached as the wave is made more intense (Fig. 6).

An honest statement about the operating point. The two corrections above have opposite signs, and at $I_0 = 0.7$ they partially cancel. This is visible in Fig. 3, where the emulation points lie between the first-order NLSE line and the exact classical curve. We chose the operating point before the sweep, on the basis of exploratory scans recorded in `attempts_log.md`, and we report the full intensity dependence so that the cancellation is visible rather than hidden. The mapping itself — Eq. (13) — is verified independently and to machine precision against the exact classical dispersion relation (3.6×10^{-5}); what the I_0 -dependence shows is the accuracy with which a 12-qubit emulation can be expected to reproduce a classical field prediction, which is a property of the lattice size, not of the mapping.

7.3 Small coupling: the ponderomotive shift sinks below the fluctuation floor

The two failed sweep points are both at small ϵ . The reason is visible in Eq. (22): the double difference subtracts two $\mathcal{O}(0.1-1)$ frequencies to expose a shift that, at $\epsilon = 0.5$, is only $0.16 J/\hbar$ in magnitude.

The residual drift left by the finite-quantum-number correction does not cancel exactly between the extended and unextended runs, because the carrier itself changes slightly with ϵ (the Bloch solution of Eq. (10) depends on ϵ), and this residual sets a floor of order $0.07 J/\hbar$, which is precisely the discrepancy seen at $\epsilon = 0.5$. Empirically the measurement is reliable once $|q_{zz}|I_0 \gtrsim 0.4 J/\hbar$, i.e. $\epsilon \gtrsim 1.2$ at this operating point.

7.4 The classical instability itself is not realised at twelve qubits

The most important limitation is one we can state quantitatively. The nonlinear coefficient measured here predicts, through Eq. (16), a modulational instability with $\Gamma(K_1) = 0.2580$ at $\epsilon = 2$, and the classical sector of the same Hamiltonian reproduces that rate to 3.4% or better. The full quantum emulation does not: seeding the same modulation on the 2^{12} -dimensional statevector and tracking its linear response, we find no exponential growth at any point of the sweep, at any carrier intensity we can reach.

The reason is a competition of timescales. Mean-field dynamics is exact for a product state at $t = 0$ and departs from it as connected correlations grow, on a timescale that lengthens with the number of quanta. At 12 sites the carrier coherence $|P(t)|/|P(0)|$ has fallen to 0.57 under H_0 and to 0.11 under $H_0 + H_{ZZ}$ at $\epsilon = 2$ by the end of the measurement window $t = 4 \hbar/J$ (Fig. 4, right), whereas the instability needs $1/\Gamma \simeq 4 \hbar/J$ per e-folding and several e-foldings to emerge from a seed. The two windows are comparable, so the instability is destroyed by quantum fluctuations before it can develop. The nonlinear frequency shift, by contrast, is a *first-order* effect in the interaction and is fully accumulated within the coherent window, which is why it survives.

This is a statement about lattice size, and it is quantitative: the coherent window scales with the number of quanta $M \propto N_s$, while $1/\Gamma$ is N_s -independent at fixed intensity, so the ratio improves linearly in N_s . Reaching several e-foldings inside the coherent window requires $M \gtrsim 20$ quanta and hence $N_s \gtrsim 40$ sites — a 2^{40} -dimensional statevector, which needs 17.6 TB of memory and is therefore out of reach of exact classical emulation (Sec. 8), and precisely the regime in which a quantum processor would be the natural instrument. We regard this as the sharpest quantitative statement in the paper about where the quantum advantage for nonlinear plasma dynamics begins.

8 Computational cost and where the advantage lies

All timings are single-core, on the machine described in `proposal/codebase_notes.md`, and are logged in `proposal/logs/controls.json`.

Emulation. One measurement run — state preparation plus $T/\delta = 4/0.2$ Trotter steps on the full 2^N -dimensional statevector, including the site-resolved observables — costs 14 ms at $N = 9$, 16 ms at $N = 10$, 21 ms at $N = 11$ and 29 ms at $N = 12$. The growth is the expected $\mathcal{O}(N 2^N)$ of gate-by-gate statevector propagation: a factor 2.2 from $N = 9$ to $N = 12$ against a Hilbert-space growth of 8. The complete seven-point sweep, including the two infinitesimal-amplitude reference runs per point, takes 1.6 s.

PDE reference. Evaluating Eq. (16) and Eq. (23) costs 0.008 ms per point — these are closed forms. Integrating the full classical field equations (6)–(7) on the same lattice, which is the reference used for the modulational-instability comparison, costs 59 ms for a 121-point trajectory. The classical reference is thus three to four orders of magnitude cheaper than the emulation at $N = 12$, which is the expected situation: at this size there is no advantage of any kind, and none is claimed.

Scaling and the crossover. The relevant comparison is not emulation-versus-PDE but classical-versus-quantum evolution of the same Hamiltonian. For H_0 the model is a free lattice field theory: the single-particle problem is tridiagonal, so a wave-packet evolution costs $\mathcal{O}(K N_s^2)$ and a hundred-site lattice is trivial. Adding H_{ZZ} removes that structure. Under the Jordan–Wigner transformation H_0 maps to free fermions while H_{ZZ} becomes a nearest-neighbour density–density interaction, so the model becomes the interacting XXZ chain; its time evolution from a product state with finite energy density is not computable in polynomial time by any known general method, and exact statevector propagation costs $\mathcal{O}(N_s 2^{N_s})$ per Trotter step with $\mathcal{O}(2^{N_s})$ memory. Our measured 1.5 ms per Trotter step at $N_s = 12$ (0.07 MB of state) extrapolates to 365 h per Trotter step and 17.6 TB of state at

$N_s = 40$; even $N_s = 36$ already needs 1.10 TB, so exact statevector emulation of the extended model runs out well before the lattice is large enough to host the physics (Sec. 7).

The quantum cost is unchanged by the extension. A Trotter step of $H_0 + H_{ZZ}$ has exactly the same two-qubit gate count as a Trotter step of H_0 , because the ZZ layer is the ϕ angle of the entangling gate that is being applied anyway; the circuit depth remains $\mathcal{O}(1)$ per step and $\mathcal{O}(N_t)$ overall, with $N_t \propto N_s$ from the Courant–Friedrichs–Lewy condition. The extension therefore converts a problem with a polynomial classical algorithm into one with an exponential classical cost at zero quantum cost, which is the sense in which longitudinal couplings are the natural route to advantage for this family of models.

Section 7 sharpens where that advantage would first be scientifically useful rather than merely formal: the modulational instability this work characterises requires $N_s \gtrsim 40$ sites before it can develop inside the coherent window, which is exactly where classical statevector emulation stops being comfortable. We note that matrix-product-state methods would extend the classical reach for short times, but the instability under study generates entanglement precisely because it converts a product state into a correlated one, so the bond dimension required grows with the same exponent that governs the instability.

9 Relation to existing work

Nonlinear plasma wave theory. The envelope equation we arrive at is the ponderomotive nonlinear Schrödinger equation of plasma physics. Its derivation from an intensity-dependent dispersion relation is due to Ostrovskii [Ostrovskii, 1967] and Karpman and Krushkal [Karpman and Krushkal, 1969]; the modulational criterion is Lighthill’s [Lighthill, 1965], with the deep-water analogue of Benjamin and Feir [Benjamin and Feir, 1967], and the plasma-specific analysis including Langmuir collapse is Zakharov’s [Zakharov, 1972, Zakharov and Shabat, 1972]. The specific nonlinearity we generate — a plasma frequency depressed in proportion to the local intensity — is the one identified by Kaw and Dawson [Kaw and Dawson, 1970] and developed by Max, Arons and Langdon [Max et al., 1974], Shukla *et al.* [Shukla et al., 1986] and Sun *et al.* [Sun et al., 1987]; textbook treatments are in Kruer [Kruer, 1988] and Whitham [Whitham, 1974]. Our contribution is not to these results, which we use as the reference, but to showing that a nearest-neighbour longitudinal spin coupling reproduces them with a coefficient computable in closed form.

Spin chains and the NLSE. That spin systems and the NLSE are connected is classical: the continuum Heisenberg ferromagnet is gauge-equivalent to the focusing NLSE [Lakshmanan, 1977, Zakharov and Takhtajan, 1979]. Our reduction is different in an important respect and we want to be explicit about it. The Lakshmanan–Zakharov–Takhtajan equivalence is an exact statement about the isotropic continuum ferromagnet; what we do is the more pedestrian spin-coherent-state (time-dependent variational) reduction, whose lattice form is the discrete NLSE familiar from Bose–Einstein condensates in optical lattices [Trombettoni and Smerzi, 2001, Smerzi et al., 2002, Kevrekidis, 2009], followed by an envelope expansion. The Holstein–Primakoff route [Holstein and Primakoff, 1940] would give the same answer to this order; we use the hard-core (Matsubara–Matsuda) variables [Matsubara and Matsuda, 1956] because they are exact for spin- $\frac{1}{2}$ and make the phase-space-filling nonlinearity explicit rather than an $\mathcal{O}(1/S)$ correction. The modulational instability of the discrete NLSE has been analysed in detail [Kivshar and Peyrard, 1992, Smerzi et al., 2002] and observed in atomic condensates [Strecker et al., 2002, Nguyen et al., 2017]; what is new here is the identification of the lattice coefficients with plasma quantities and the sign competition between the hard-core and ZZ nonlinearities. We claim no novelty for the reduction technique itself.

Quantum simulation of plasma physics. Algorithms for plasma-relevant equations now cover wave propagation, advection and kinetics [Engel et al., 2019, Joseph, 2020, Dodin and Startsev, 2021, Novikau et al., 2022, Koukoutsis et al., 2023], and the field has been surveyed [Amaro and Cruz, 2023]. Most of this work targets linearised classical equations, where quantum advantage is hard to establish [Dodin and Startsev, 2021]; the exceptions target intrinsically quantum dynamics [Shi et al., 2021]. The spin-chain plasma-wave model [Sundar et al., 2026] is attractive because it is the shallowest such construction, and it identifies longitudinal couplings as the route to nonlinearity but leaves the corresponding plasma physics open. This paper closes that gap for one coupling.

XY chains with longitudinal couplings on hardware. The Hamiltonian we study is, in fermionic language, an XXZ chain with a staggered field, and XY -plus- ZZ chains are among the most heavily studied models on superconducting processors; the formation of bound states of interacting microwave photons in an FSIM chain [Morvan et al., 2022] is the same Hamiltonian family in a different physical language, and the correspondence between the two-quantum bound state there and the bright soliton of our NLSE limit is exact. We regard this convergence as a strength: the hardware requirements of the plasma application have already been demonstrated in a condensed-matter framing, and what has been missing is the dictionary. Error mitigation for exactly this circuit family — pseudotwirling of the non-Clifford FSIM gate, Clifford data regression, classical shadows [Wallman and Emerson, 2016, Czarnik et al., 2021, Huang et al., 2020, Foxen et al., 2020] — is established.

10 Limitations and outlook

What we established. A uniform nearest-neighbour $\sigma^z\sigma^z$ coupling added to the staggered-mass XY chain generates the ponderomotive nonlinearity of plasma waves. The identification is analytic: the coupling appears in the classical field equations solely as an intensity-dependent plasma frequency, $\Delta_{\text{eff}} = \Delta + \epsilon J(1 - 2\bar{n})$, and the resulting envelope equation is the cubic NLSE with $q_{zz} = 2\epsilon J r_0^2 / (1 + r_0^2)^2$. Evolving the full 2^{12} -dimensional Hilbert space and measuring the ponderomotive frequency shift interferometrically reproduces this coefficient to within the pre-registered 15% tolerance at 5 of 7 declared sweep points across an 8-fold range of coupling, while the identical protocol inside the single-excitation subspace returns zero to 2.1×10^{-3} . In the classical sector of the same Hamiltonian the implied modulational growth rate matches the NLSE to 3.4% or better, and the Lighthill sign change predicted at $\epsilon_c = 1.031$ is confirmed.

What the model can be expected to capture. Any nonlinear plasma process that reduces, in the weakly nonlinear limit, to a local intensity-dependent refractive index: modulational instability and its saturation, self-phase modulation, envelope soliton formation, and — with a spatially varying Δ_j , which the linear model already supports — ponderomotive channelling and self-induced transparency of an overdense layer. The coefficients are all available in closed form from Eq. (13), so predictions can be made before running.

What it cannot. Three limits are structural rather than technical. (i) The nonlinearity is *local*. Processes governed by a nonlocal response — Debye screening at finite temperature, the Zakharov coupling to a separately evolving ion-acoustic field, wakefield generation — require either a longer-range K_{ij} or a second species and are outside this model. (ii) The hard-core constraint contributes its own nonlinearity, q_{psf} , which cannot be switched off. It is subdominant at the couplings we use but it means the model is never purely ponderomotive; quantitatively it sets the threshold ϵ_c . (iii) Most importantly, at 9–12 qubits the emulation is not a classical field. It contains 2.85 wave quanta at our operating point, enough to accumulate a first-order nonlinear phase inside the coherent window but not enough to grow the instability that phase implies. We showed this by direct simulation rather than asserting it, and we estimated the lattice size at which it changes ($N_s \gtrsim 40$).

Outlook. The immediate experimental step is small: the circuits are those already run for the linear demonstration with the FSIM ϕ angle retuned, the state preparation is one single-qubit rotation per qubit, and the observable is the same $\langle \sigma_j^x \rangle, \langle \sigma_j^y \rangle$ set, measured by classical shadows. The double-difference structure of Eq. (22) is well suited to a noisy device because the four runs share a circuit and differ only in single-qubit preparation angles, so depolarising suppression largely cancels. The scientific step is larger and is where the model becomes worth a quantum computer: at forty sites the same Hamiltonian supports a genuine modulational instability, its saturation into envelope solitons, and the onset of weak turbulence, none of which is accessible to exact classical statevector emulation — and all of which have a quantitative classical-PDE prediction, derived here, to test against.

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A Derivation details

A.1 Effective fields

With $H = \sum_j \vec{B}_j \cdot \vec{\sigma}_j$ and $E = \langle H \rangle$ evaluated on a product state, the effective fields of $H_0 + H_{ZZ}$ are

$$\begin{aligned} B_j^x &= \frac{J}{4} (s_{j-1}^y - s_{j+1}^y), & B_j^y &= \frac{J}{4} (s_{j+1}^x - s_{j-1}^x), \\ B_j^z &= \frac{1}{2} \Delta_j (-1)^j - \frac{\epsilon J}{4} (s_{j-1}^z + s_{j+1}^z). \end{aligned} \quad (24)$$

The combination that appears in the equation of motion for $\sigma_j = s_j^x + i s_j^y$ is $B_j^y - i B_j^x = \frac{J}{4} (\sigma_{j+1} - \sigma_{j-1})$, which is what makes Eq. (6) close.

A.2 Solving the amplitude equation to first order in intensity

Setting $\mu = -\omega + \epsilon J$, $A_e = a$, $A_o = ra$ and $z_s = \sqrt{1 - A_s^2} \simeq 1 - \frac{1}{2} A_s^2$ in Eq. (10) (units $J = 1$) gives the pair

$$\mu = r \sin k + \Delta - \frac{a^2}{2} (r \sin k + \epsilon r^2), \quad \mu = \frac{\sin k}{r} - \Delta - \frac{a^2}{2} (r \sin k + \epsilon). \quad (25)$$

Their difference is the amplitude equation $F(r) \equiv r \sin k + 2\Delta - \sin k/r = \frac{a^2}{2}\epsilon (r^2 - 1)$, whose $a = 0$ root is $r_0 = (\mu_0 - \Delta)/\sin k$ with $\mu_0 = \sqrt{\Delta^2 + \sin^2 k}$. Writing $r = r_0 + r_1 a^2$ and using $F'(r_0) = \sin k (1 + r_0^{-2})$ gives

$$r_1 = \frac{\epsilon r_0^2 (r_0^2 - 1)}{2 \sin k (r_0^2 + 1)}, \quad \mu_1 = r_1 \sin k - \frac{r_0 \sin k}{2} - \frac{\epsilon r_0^2}{2} = -\frac{r_0 \sin k}{2} - \frac{\epsilon r_0^2}{1 + r_0^2}. \quad (26)$$

With $I = \frac{1}{2}a^2(1 + r_0^2)$ and $\Omega = -\mu + \epsilon J$ one obtains $q = 2\mu_1/(1 + r_0^2)$, which is Eq. (13). The full symbolic derivation, including the intermediate expressions, is in `proposal/logs/derivation.txt`.

B Verification suite

`proposal/code/tests.py` writes `proposal/logs/tests.json`. All seven tests pass.

Test	What is checked	Result
T1a	each Trotter layer equals an independently formed $e^{-i\delta h}$	2.2×10^{-16}
T1b	composed step converges to $e^{-i\delta H}$; measured local order	2.00
T2	norm of the 2^N -dimensional statevector under evolution	6.6×10^{-14}
T3	classical field equations = exact Heisenberg equations for product states	4.4×10^{-16}
T4	finite-amplitude Bloch carrier is a fixed point of the co-rotating flow	5.0×10^{-11}
T5	analytic q, p against numerical derivatives of the exact $\Omega(k, I)$	$3.6 \times 10^{-5}, 5.2 \times 10^{-8}$
T6	$H_{ZZ}(0) = 0$; ZZ path $\equiv$ unextended path bit-for-bit	0.0
T7	evolution occupies excitation sectors outside the initial one	13 sectors

C Log index

Every number in this paper is generated by `proposal/code/make_macros.py` from the following logs and inserted as a macro; none is typed by hand.

File	Contents
<code>logs/method_plan_committed_at.txt</code>	timestamp of the pre-registration
<code>logs/derivation.txt</code>	symbolic derivation D1–D5 with intermediates
<code>logs/derivation_coefficients.json</code>	closed forms consumed by the numerics
<code>logs/tests.json</code>	verification suite T1–T7
<code>logs/linear_limit.json</code>	$N = 9$ dispersion and propagation, $\epsilon = 0$ equivalence
<code>logs/sweep.json</code>	headline sweep, Trotter convergence, intensity scan
<code>logs/classical_mi.json</code>	modulational growth rates, $\Gamma(K)$, Lighthill scan
<code>logs/controls.json</code>	linear limit, truncation control, Hilbert-space diagnostics, timings
<code>logs/contrast.json</code>	H_0 contrast trajectories at $\epsilon = 2$
<code>logs/h0_baseline.json</code>	unextended- H_0 runs at every declared sweep point
<code>logs/trajectories.npz</code>	every amplitude of the 2^{12} statevector at every recorded step, for all sweep points
<code>logs/macros.tex</code>	the generated macros